ANKIT SHARMA

+91 8115080798 ◇ Gorakhpur, Uttar Pradesh, India
ankit972125@gmail.com ◇ LinkedIn ◇ GitHub ◇ Portfolio

PROFESSIONAL SUMMARY

Software Engineer with 1+ years of professional experience in designing, developing, and maintaining scalable software systems and web applications. Experienced in distributed system architectures, backend and frontend development, and performance optimization. Strong foundation in data structures, algorithms, system design, and DevOps practices, with a proven track record of delivering reliable, maintainable, and production-ready solutions that improve system efficiency and overall application performance.

EDUCATION

Degree / Certificate	Institute / Board	CGPA / %	Year
B.Tech (Information Technology)	Rajkiya Engineering College, Azamgarh	8.06 CGPA	2022 – 2026
12th (Intermediate)	UP Board	76.8%	2022
10th (High School)	UP Board	85.1%	2020

Relevant Coursework: Data Structures & Algorithms, Computer Networks, Operating Systems, Machine Learning, Deep Learning, Natural Language Processing, DBMS, Software Engineering, Compiler Design

Academic Projects: Distributed File System, Neural Machine Translation System, Real-time Chat Application

CERTIFICATIONS & TRAINING

- **Problem Solving (Basic)** – HackerRank (DSA Algorithms) 2024
- **Cyber Security Fundamentals** – Simplilearn 2024
- **AWS S3 Basics** – Amazon Web Services (Cloud Storage IAM) 2024
- **Web Development Workshop** – Microsoft (Frontend Web Fundamentals) 2024

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript (ES6+), Go, Java, C/C++, SQL, Bash/Shell
AI/ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Keras, Hugging Face Transformers, NLTK, spaCy
LLMs & Generative AI: OpenAI API, Anthropic Claude, LangChain, LlamaIndex, RAG Architectures, Prompt Engineering, Fine-tuning, Quantization
Vector Databases & Search: Pinecone, ChromaDB, FAISS, Weaviate, Elasticsearch
Backend Development: FastAPI, Django, Flask, Node.js, Express.js, Fiber (Go), Gorilla Mux, Celery, Socket.io
Frontend Development: React.js, Next.js, Redux Toolkit, Zustand, Tailwind CSS, Material-UI, HTML5, CSS3, Bootstrap, Angular
APIs & Microservices: REST APIs, GraphQL, gRPC, Protocol Buffers, WebSockets, JWT/OAuth2, Kafka (Sarama)
Database Systems: PostgreSQL, MongoDB, MySQL, Redis, SQLite, Supabase, Firebase
Cloud & DevOps: Unix/Linux, AWS (EC2, S3, RDS, Lambda, CloudFront), Docker, Kubernetes, GitHub Actions, CI/CD, Nginx, Vercel, Railway
Architecture Patterns: Microservices, Event-Driven Architecture, SAGA Pattern, Clean Architecture, Distributed Systems, CQRS
Automation & Tools: n8n, Selenium, FireCrawl, Jina, Beautiful Soup, Scrapy, Git, Postman, Jira, Figma

PROFESSIONAL EXPERIENCE

Systems Engineer June 2026 – Present
Tata Consultancy Services (TCS), Mumbai

- Working with Java, Spring Boot, SQL, and Git to develop and maintain enterprise applications
- Collaborating with team members to understand business requirements, resolve application issues, and contribute to feature development

- Following Agile methodologies, coding standards, and best practices while delivering high-quality software solutions

Python Developer Intern January 2026 – June 2026
ShortOrbit Technologies Pvt Ltd

- **Zivoro**
  - Maintained and enhanced production-grade backend systems using Django and FastAPI, ensuring high system stability and reliability
  - Diagnosed and resolved critical production issues, reducing system downtime and improving overall service availability
  - Optimized REST APIs and database queries, resulting in improved response times and backend performance
  - Collaborated with frontend team (Angular) to deliver seamless feature integrations and improve user experience
  - Contributed to code quality improvements, refactoring legacy components and following best engineering practices

Software Developer Intern November 2025 – January 2026
Cruv

- **CaliberOS — AI Operating System for Insurance**
  - Engineered a scalable file storage system similar to Google Drive with secure upload, access control, and file management capabilities
  - Developed a full-featured email service supporting sending, receiving, and real-time notifications, mimicking core Gmail functionalities
  - Designed and implemented robust REST APIs using Django REST Framework for handling complex backend operations
  - Built efficient backend logic for user interactions, ensuring data consistency and optimized performance
  - Improved system scalability and maintainability through modular architecture and clean code practices

Junior Software Engineer (AI/ML Focus) April 2025 – January 2026
Meteorite Labs

- **SeaMaster-AI — AI-Powered Information Retrieval Platform**
  - Designed and developed a large-scale AI-powered information retrieval system using Python (Flask), Next.js, and Retrieval-Augmented Generation (RAG), handling 10,000+ queries per day
  - Implemented semantic search using Pinecone vector database and custom embedding models, improving answer relevance by 75% and reducing query latency to under 200ms
  - Improved LLM throughput by 60% through Redis-based caching, prompt optimization, and sandboxed code execution
  - Integrated Zoho Subscriptions for plan-based access control, usage tracking, and automated billing workflows
  - Built robust LLM guardrails using GuardrailsAI, reducing hallucinations and invalid outputs by 85%
  - Enabled multi-LLM support (GPT-4, Claude, Llama) with intelligent fallback and failover mechanisms
- **LocalQR — QR-Based Personalized Recommendation Platform**
  - Developed a personalized QR-based recommendation and guide-sharing platform focused on frontend architecture and user experience
  - Built responsive and accessible UI using modern frontend frameworks, ensuring seamless cross-device compatibility
  - Integrated REST APIs for dynamic content delivery and real-time personalization
  - Optimized frontend performance through efficient state management and rendering strategies, improving load times and user interaction responsiveness

- **LocalTunes — Real-Time Cross-Platform Music Streaming Platform**
  - Contributed to the development of a real-time, cross-platform music streaming platform allowing users to request and queue music dynamically
  - Implemented real-time request handling and synchronization for live music playback scenarios
  - Integrated subscription management features including plan enforcement, access control, and usage limits
  - Collaborated on backend and frontend integration to ensure low-latency interactions and consistent user experience across platforms

Software Developer Intern (Frontend) November 2024 – May 2025
Hirreetech Consulting and Services Certificate

- **Hirree — Hiring and Talent Discovery Platform**
  - Developed and maintained production-grade frontend features for Hirree, a large-scale hiring and recruitment platform connecting candidates and employers
  - Built modular and reusable UI components using React, Redux Toolkit, and Tailwind CSS with seamless REST API integration
  - Improved platform stability by reducing frontend-related downtime by 20% through systematic debugging, performance profiling, and implementation of React error boundaries
  - Enhanced user engagement by 15% through responsive design improvements, AI-assisted UI personalization, and cross-device compatibility testing
  - Optimized state management and component rendering logic, reducing page load times by 35%
  - Collaborated closely with backend, product, and design teams in Agile sprints, participating in code reviews, sprint planning, and daily standups

Django Developer Intern (Freelance Projects) May 2024 – September 2024
Fox Global Technologies Certificate

- **Hospital Management System (Django)**
  - Developed a full-featured hospital management system using Django and Django REST Framework
  - Implemented modules for patient records, appointments, staff management, billing, and role-based access control
  - Optimized database queries and introduced caching, reducing API response times by 20%
  - Integrated secure authentication, audit logs, and admin dashboards for operational monitoring
- **Chess Game Engine and Interface (Python)**
  - Built a chess application using Python and the PyChess ecosystem with rule validation and game state management
  - Implemented move validation, game history tracking, and interactive board rendering
  - Designed modular code structure to support AI-based move analysis and future enhancements
- **Project Management System (Django)**
  - Developed a project management platform using Django and REST APIs supporting task tracking, deadlines, and team collaboration
  - Implemented role-based permissions, activity logs, and real-time status updates
  - Designed RESTful APIs with Swagger/OpenAPI documentation for frontend and third-party integrations
- **CRM and Business Automation Systems**
  - Delivered multiple CRM solutions for different clients, including lead management, customer pipelines, and reporting dashboards
  - Integrated background task processing using Celery and Redis for notifications, reports, and scheduled jobs
  - Implemented data preprocessing pipelines using pandas and NumPy for analytics and ML-driven insights

PROJECTS

AI/ML & Generative AI Projects

SeaMaster-AI – AI-Powered Information Retrieval Platform GitHub

- Built a production-grade RAG-based AI system handling **10,000+ queries/day** using Flask and Next.js
- Implemented semantic search with Pinecone vector DB and custom embeddings, reducing latency to **¡200ms**
- Improved LLM throughput by **60%** via Redis caching, prompt optimization, and multi-LLM fallback
- Added hallucination guardrails using GuardrailsAI, reducing invalid outputs by **85%**

Facial Recognition System – Computer Vision Application GitHub

- Developed real-time facial detection and recognition pipeline with modern web interface
- Implemented deep learning models for accurate feature extraction and identity matching
- Integrated AI inference with responsive frontend visualization and user management
- Demonstrates end-to-end deployment of applied computer vision systems

Systems & Distributed Architecture Projects

DistriShop – Distributed E-commerce Microservices Platform GitHub

- Architected production-grade microservices system in Go with isolated user, product, and order services
- Implemented gRPC (HTTP/2) communication and unified GraphQL API gateway with type-safe resolvers
- Built Docker-based infrastructure with service discovery, health checks, Nginx load balancing, and graceful shutdowns
- Integrated recommendation engine and A/B testing framework with MLflow monitoring

AsyncFlow – Event-Driven Kafka Microservices GitHub

- Designed high-throughput event-driven architecture using Kafka with Go-based producers and consumers
- Processed **1000+ messages/second** with retries, DLQs, idempotent handling, and consumer rebalancing
- Deployed Dockerized Kafka cluster with Zookeeper, structured logging, and Prometheus observability
- Demonstrates strong expertise in distributed systems and reliability engineering

Atomic-Saga – Distributed Transaction Management Library (OSS) GitHub, NPM

- Published open-source TypeScript library implementing SAGA orchestration with **500+ weekly NPM downloads**
- Implemented compensating transactions, rollback strategies, and transactional outbox pattern
- Added idempotent API handling, workflow orchestration, and structured logging
- Authored extensive documentation with integration guides and best practices

IndexGo – High-Performance Indexing Engine GitHub

- Developed high-performance indexing and lookup utilities in Go
- Optimized memory usage and retrieval speed using custom data structures
- Demonstrates strong command over algorithms and low-level backend optimization

Backend & Infrastructure Projects

Hospital Management System – Enterprise Django Platform GitHub

- Built enterprise-grade hospital management system using Django, DRF, and PostgreSQL
- Implemented patient records, appointments, billing, RBAC, and admin dashboards
- Optimized queries and caching, reducing API response times by **20%**
- Integrated secure authentication, audit logs, and operational monitoring

Pokemon-CLI – High-Performance Go CLI Tool GitHub

- Built concurrent Go-based CLI tool consuming REST APIs for Pokémon data
- Reduced API calls by **80%** using Redis caching and improved response speed by **3×**
- Designed interactive terminal UI with search, filtering, pagination, and detailed statistics
- Achieved **90% code coverage** with robust unit tests and error handling

LocalBase – Lightweight Embedded Database GitHub

- Designed file-based embedded storage system for efficient local persistence
- Implemented optimized read/write operations, indexing, and query processing
- Demonstrates storage system design beyond traditional CRUD applications

go-file-encryption – Secure Systems Utility GitHub

- Implemented secure file encryption/decryption using Go cryptographic libraries
- Followed industry best practices for key handling and data protection
- Highlights security-first mindset in backend and infrastructure development

Full-Stack & Real-Time Applications

Code with Friend – Real-Time Collaborative Coding Platform GitHub, Demo

- Built real-time collaborative editor using React, Socket.io, and Django Channels
- Supported **20+ concurrent users** with sub-**100ms** synchronization latency
- Implemented WebSocket sync, Monaco editor, live cursors, JWT auth, and RBAC
- Deployed on AWS EC2 with Nginx and horizontal scaling via Redis pub/sub

StudySphere – Educational Management System GitHub, Demo

- Developed full-stack education platform serving **500+ active users**
- Implemented JWT-based RBAC for admins, teachers, and students
- Automated attendance, grading, assignments, and notifications, reducing workload by **70%**
- Optimized performance with caching and indexing, reducing page load times by **40%**

CRM & Business Automation Systems – Client Projects GitHub

- Delivered multiple CRM platforms for lead management, pipelines, and analytics
- Integrated Celery and Redis for background processing and scheduled workflows
- Built analytics pipelines using pandas and NumPy for data-driven insights

Frontend & Product Applications

Developer Portfolio Platform Live, GitHub

- Built modern, responsive portfolio using React and Next.js
- Optimized for performance, SEO, lazy loading, and accessibility
- Serves as a central personal branding and project showcase

Product-Oriented Web Applications GitHub

- Developed multiple frontend apps (travel-norway, teenhive, huzzle, WhatBytes)
- Focused on modular UI, accessibility, and responsive design
- Integrated REST APIs for dynamic, real-time user experiences

OPEN SOURCE CONTRIBUTIONS

- **LangChain:** Contributed documentation improvements and bug fixes for vector store integrations
- **React-Redux:** Submitted pull request optimizing middleware performance in production builds
- **FastAPI:** Added examples for WebSocket integration patterns and async database operations
- **Community Engagement:** Active contributor on Stack Overflow (1500+ reputation), helping developers with Python, React, and system design questions

ACHIEVEMENTS & RECOGNITION

- Solved 1000+ algorithmic problems across LeetCode (600+), GeeksforGeeks (250+), and HackerRank (150+) with focus on data structures, algorithms, and system design
- Achieved 5-star rating in C/C++ and 3-star rating in Python on HackerRank demonstrating strong problem-solving skills
- Published open-source NPM library (atomic-saga) with 500+ weekly downloads, comprehensive documentation, and active community support
- Successfully delivered 5+ freelance full-stack applications generating $1,000+ in revenue with 100% client satisfaction
- Winner of College Hackathon 2024 for building AI-powered study assistant using RAG and fine-tuned language models

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Technical Lead** – College Coding Club 2024 - Present
  - Organized 15+ technical workshops on web development, AI/ML, and competitive programming for 200+ students
  - Mentored 30+ junior students in full-stack development and helped them secure internships
- **Open Source Advocate** – Conducted sessions on Git, GitHub workflows, and contributing to open-source projects
- **Volunteer** – NGO Tech Initiative, teaching programming basics to underprivileged students (50+ hours)